Path Texture and the Cross-Section of Indian Equity Returns: Permutation Entropy and the Variance Ratio

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Abstract

Two measures of the *shape* of a stock's recent daily return path were tested as cross-sectional return predictors on a snapshot-based approximation to the Nifty 500: normalized permutation entropy (order 4) and the 21-day Lo–MacKinlay variance ratio, both over a 232-row window. The variance ratio stood in for a planned wavelet measure that was never built. Each test had a registered first gate. Neither measure passed it: Fama–MacBeth Newey–West t-statistics were -0.29 (entropy) and -0.41 (variance ratio) even before controls, and all 95% intervals include zero. A revision diagnostic finds that most stock-months have entropy close to the finite-sample i.i.d. benchmark. Simulated noise of the same length reproduces about two-thirds of the cross-sectional entropy variance. The entropy test used 2011–2026 data including 2022 onward; a post hoc 2011–2021 re-estimate also fails the gate.

Read before interpreting results.

- **Sample.** Test M7 (entropy) used monthly rebalances 2011-01 to 2026-06, including 2022 onward. Tests D (variance ratio) and C (appendix) used 2010–2021 only; Test D's last forward window reads 21 trading days of January 2022, and a strict re-estimate that drops it is reported. No result here is validated on untouched data.
- **Multiple testing.** Legacy ledger: M7 is item 29 (the momentum-enhancement batch's internal count, Sahi [6]); the Test C gate and its diagnostics are 30 and 31; the Test D gate is 32. The revision diagnostics in Sections 5–9 are post hoc, are not registered tests and are not in that count. No multiplicity adjustment was applied because no test passed.
- **Costs.** No portfolio was formed. All statistics are gross cross-sectional coefficients or correlations.
- **Point-in-time universe.** Membership is a snapshot-based point-in-time approximation built from semi-annual Nifty 500 lists; the 2018-06-30 snapshot is truncated (about 353 instead of about 500 names, carried forward from 2017-12-31), so up to about 22% of names may be misclassified in July–December 2018.
- **Other.** No wavelet transform was computed. Seven development-period month-ends are absent or dropped (Section 2); a calendar-corrected rerun is pending and could change sample counts and estimates slightly. All 95% intervals were added during drafting or revision.

1 Introduction

Momentum ranks stocks by the *magnitude* of their past return. A natural follow-up question is whether the *texture* of the path also matters: whether stocks reached the same 12-month return smoothly or in jumps, and whether their movement was concentrated at slow or fast horizons. The "frog-in-the-pan" result of Da et al. [2] is the best-known version of this idea. Permutation entropy [1] and multiscale variance measures are model-free ways to describe path shape.

This paper reports two gated tests of the idea on Indian equities. Test M7, run inside a momentum-enhancement batch [6], asked whether, at equal 12–1 momentum, stocks with low

permutation entropy (smooth paths) later outperformed stocks with high entropy (jumpy paths). Test D asked whether the balance between slow-scale and fast-scale movement, measured by the variance ratio, predicted forward returns independently of volatility. In both tests, cheap checks had to pass before any portfolio sort was allowed.

Contribution. For the two specified measures, on daily returns over a 232-row window, neither passed its registered first forward-return gate. The 95% intervals rule out large effects more clearly than small ones. The diagnostics show that the entropy measure is close to its finite-sample i.i.d. reference for most stock-months, and that much of the cross-sectional variation in either measure is of the size that sampling noise alone produces. This is consistent with measurement noise being a substantial part of the story, but it does not prove that noise is the only reason for the null. The paper is complementary to Sahi [5], which studies regime-dependent momentum and reversal ICs and does not use path-shape measures.

2 Data and universe

All tests used one daily panel (1,069 tickers, 2010-01-04 to 2026-07-10; file `feature_panel_with_pit_v2_2026-07-24`), documented in https://github.com/mriganksahi-beep/India-Equity-Research/tree/main/data_pipeline. Daily returns are close-to-close on the ticker's prior trading row. Placeholder rows that the data vendor creates on non-trading days were masked first. `realized_vol_60` is the annualized 60-row standard deviation of daily returns, turnover is volume × close, and Amihud illiquidity is a monthly mean of $|r|$/turnover.

A rebalance date t is the last row of each calendar month in the panel's date index. A stock is eligible if its membership flag equals 1 on t, it is not one of eight tickers with known corporate-action splices, it has a complete 12–1 momentum window, and at least 200 of the 232 rows from $t-252$ to $t-21$ (inclusive) have a return. The 252-row lookback with a 21-row skip leaves this 232-row window, and both path measures use only it. The forward return compounds rows $t+1$ to $t+21$. Months with fewer than 100 eligible names are dropped.

Skipped months. Across 2011–2021, seven month-ends are absent or dropped in both the entropy and the variance-ratio panels. Five months have no rebalance row at all: 2011-08, 2014-08, 2016-10, 2018-03 and 2020-11. Their month-ends coincide with exchange holidays on which the panel's date index carried rows that most tickers left blank. Two further months are below the 100-name floor. 2015-02-28 (56 names) was a genuine Saturday session held for the Union Budget, and its cross-section is thin. 2019-03-29 (30 names) is unexplained. In the full M7 sample, 2025-03, 2025-04 and 2025-05 are also dropped because `realized_vol_60` is missing for nearly all names after a vendor-failure day on 2025-03-18. A calendar-corrected rerun of the panel is pending. All sample counts and estimates that it could change are listed in `RERUN_DEPENDENT.md`. The development slice used for M7 diagnostics, which equals the Test D panel, has 59,184 stock-months on 865 tickers.

3 Method

3.1 Permutation entropy

Each run of m consecutive returns is mapped to its ordinal pattern, the order that sorts it.[1] With p_π the share of pattern π among the $n-m+1$ windows [1],

$$\mathrm{PE}_m = -\frac{1}{\ln m!}\sum_{\pi} p_\pi \ln p_\pi \in [0,1]. \tag{1}$$

[1] For $m=4$, the returns $(0.2\%, -1.0\%, 0.5\%, 0.1\%)$ sort as positions $(2,4,1,3)$. That pattern is one of $4! = 24$ possible patterns.

M7 used $m = 4$, $\tau = 1$ (primary) and $m = 3$ as a robustness variant. A monotone path uses one pattern (PE = 0). For an i.i.d. continuous series, every pattern is equally likely whatever the marginal distribution. Even so, a finite series of about 229 windows has PE strictly below 1, because the 24 empirical frequencies fluctuate around 1/24. The relevant benchmark is therefore the finite-sample i.i.d. distribution, not 1. Ties (repeated values, such as zero-return days) are ordered by `numpy.argsort`'s default sort, which is deterministic but not stable, so tied values are not guaranteed to keep their time order. Section 5.2 tests sensitivity to this rule. Sample entropy [4] with $m = 2$ and $r = 0.2\hat{\sigma}$, and the DGW control (the share of positive days), use the same window.

3.2 Variance ratio as a stand-in for wavelet scale energy

Test D was motivated by a wavelet idea: the share of path energy at coarse scales. Total wavelet energy equals the sample variance, so raw band energies are volatility in disguise. The design therefore gated on a scale-free quantity that addresses a related question, the overlapping variance ratio [3]:

$$\widehat{\mathrm{VR}}(q) = \frac{\sum_{j=q}^{n} \left(X_j - X_{j-q} - q\hat{\mu}\right)^2 / \left[q(n-q+1)(1-q/n)\right]}{\sum_{t=1}^{n}(r_t - \hat{\mu})^2/(n-1)}, \qquad X_j = \sum_{t \le j} r_t, \tag{2}$$

with $q = 21$ and no further finite-sample correction. $\mathrm{VR}(q) > 1$ indicates positive serial dependence at the 21-day aggregation, and $\mathrm{VR}(q) < 1$ indicates negative dependence. The variance ratio is not a wavelet energy share. Its pre-build checks on simulated series of length 1,000 returned 1.87 for a trending process, 0.40 for a mean-reverting one and 1.03 for a random walk. A wavelet build (maximal-overlap transform, Daubechies least-asymmetric filter of length 8) was planned only if all three gates passed. It never ran, and no conclusion about wavelets follows from this paper.

3.3 Gate design and inference

Table 1 lists the gates. A gate that ran and missed its threshold *failed*. A gate that never ran because an earlier one failed is *not reached*. Gate-1 statistics are monthly Fama–MacBeth regressions of the 21-day forward return on cross-sectionally z-scored regressors. Inference uses the time series of monthly coefficients, with a Newey–West (Bartlett, lag 6) standard error. The decision used the *controlled* coefficient. The *uncontrolled* coefficient was also recorded, and its definition differs between tests. In M7 it comes from a regression on entropy and momentum, because the hypothesis is stated "at equal 12–1 return". In D it comes from a regression on VR alone. The uncontrolled coefficient separates two situations: a signal that the control explains away, and a signal that was never there.

4 Results

Table 2 reports gate 1. The "as run" rows were reproduced exactly from the cached panels with the notebooks' code. Coefficients are 21-day returns per one cross-sectional standard deviation of the signal; for example, −0.00059 means −0.059 percentage points per month.

M7. As run, the entropy coefficient was −0.00012 ($t = -0.29$) with momentum controlled, and −0.00005 ($t = -0.12$) after DGW was added, so gate 1 failed. The notebook's automatic label ("reinvented frog-in-the-pan") describes this failure inaccurately. DGW did not remove a signal, because the uncontrolled coefficient was already indistinguishable from zero. The sample includes 2022 onward. The 2011–2021 re-estimate was run after the full-sample result had been

Table 1: Registered gate chains. FM = monthly Fama–MacBeth; t = Newey–West (lag 6). Rules are as written in the notebooks before the gate cells ran.

Test	Gate	Pass rule	Outcome
M7	1 Distinctness	FM: PE_4 coefficient with momentum and DGW controls has $\|t\| > 2$ and the same sign as the uncontrolled coefficient	**failed** ($t = -0.12$)
	2 Volatility	gate 1 plus log RV60	not reached
	3 Stability	block-bootstrap rank stability > 0.6	not reached
	4 Performance	momentum $\times$ entropy double-sort	not computed
D	1 Vol-orthogonalization	FM: VR(21) residual on log RV60, with log RV60, momentum and log Amihud controls, has $\|t\| > 2$ and the same sign	**failed** ($t = -0.23$)
	2 Dispersion	mean monthly p10–p90 spread of VR(21) > 0.30	not reached
	3 Stability	block-bootstrap rank stability > 0.6	not reached
	– Wavelet build	MODWT, LA8	not built

Table 2: Gate-1 Fama–MacBeth results. T = months. All intervals are post hoc: 95%, coefficient ± 1.96 Newey–West standard errors. A circular block bootstrap of the monthly series (block 6, 10,000 draws) gives endpoints within ± 0.0002 of these. Rows marked † are post hoc sample restrictions added after the as-run result was seen.

Sample	Regressors	Coef.	NW t	T	95% interval
M7, 2011–2026 (as run)	PE, mom. (uncontrolled)	−0.00012	−0.29	176	[−0.00089, 0.00066]
	PE, mom., DGW (gate)	−0.00005	−0.12	176	[−0.00083, 0.00073]
M7, 2011–2021†	PE, mom. (uncontrolled)	+0.00039	+0.88	125	[−0.00047, 0.00124]
	PE, mom., DGW (gate)	+0.00048	+1.09	125	[−0.00038, 0.00134]
D, 2011–2021 (as run)	VR(21) (uncontrolled)	−0.00059	−0.41	125	[−0.0034, 0.0022]
	VR resid., RV, mom., Amihud (gate)	−0.00028	−0.23	125	[−0.0027, 0.0021]
D, to 2021-11†	VR(21) (uncontrolled)	−0.00066	−0.46	124	–
	VR resid., RV, mom., Amihud (gate)	−0.00034	−0.28	124	[−0.0027, 0.0021]

seen, so it is a restricted-sample sensitivity and not a confirmatory test. Its coefficients change sign ($t = 0.88$ and 1.09) but still fail the gate.

D. The uncontrolled VR coefficient was -0.00059 ($t = -0.41$), and the gate coefficient was -0.00028 ($t = -0.23$). Gate 1 failed, so gates 2 and 3 were not reached and the wavelet build did not run. Test D's last rebalance, 2021-12-31, has a forward window inside January 2022. Dropping it leaves 124 months, with an uncontrolled t of -0.46 and a gate t of -0.28, so the conclusion does not change.

Precision. Every interval in Table 2 includes zero. The data rule out effects larger than about 0.13 percentage points per month per standard deviation for entropy and about 0.3 for the variance ratio. They do not distinguish a zero effect from smaller ones. The appropriate reading is "not detected at the available precision".

5 Revision diagnostics

None of the analyses in this section was registered. Simulated nulls use fixed seeds (Table 3).

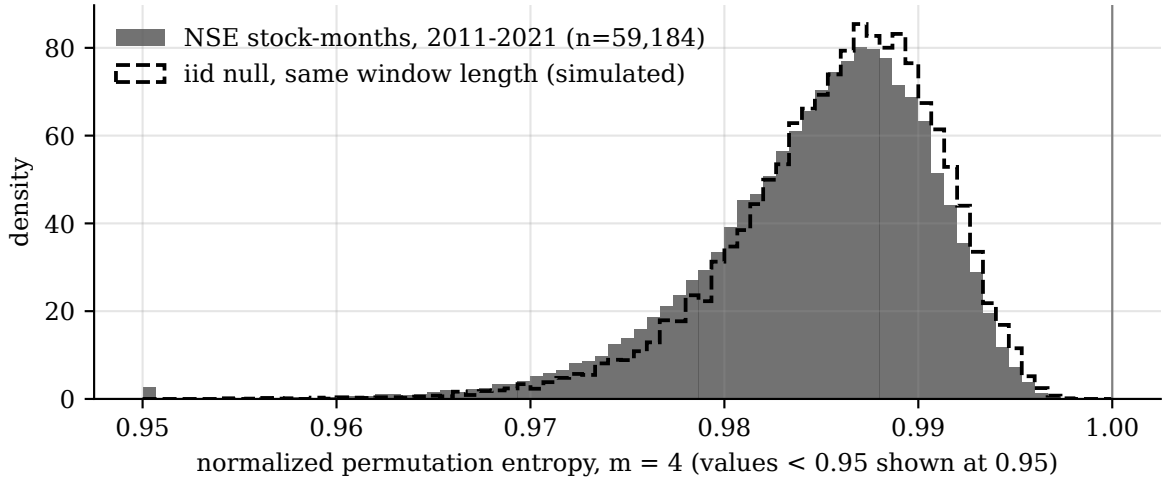


Figure 1: PE_4 for all 59,184 eligible stock-months, 2011–2021 (bars), and 20,000 simulated i.i.d. Gaussian series of length 232 (dashed). Because ordinal patterns of an i.i.d. continuous series do not depend on its distribution, the simulated null depends only on n and m. Simulated series contain no ties.

5.1 Entropy against its finite-sample i.i.d. benchmark

The 1st, 50th and 99th percentiles of PE_4 are 0.966, 0.986 and 0.995. 99.8% of stock-months are at or above 0.95, and the minimum is 0.805. The simulated i.i.d. benchmark has mean 0.986 and standard deviation 0.0053, and 98.5% of observed values lie inside its central 99% band (Figure 1). The mean within-month cross-sectional standard deviation of observed PE is 0.0065. The i.i.d. sampling variance is about 65% of the observed cross-sectional variance. So, for this estimator, window and sample, most stock-months are close to i.i.d. in ordinal terms, and most of the spread that M7 would have ranked on is of the size produced by sampling noise. This says nothing about the full return process: amplitude-based sample entropy (median 1.88) is clearly below its own i.i.d. reference (2.20, 500 draws), so the series are not i.i.d.

5.2 The low tail: ties and price discreteness

Only 96 development stock-months on 18 tickers have $PE_4 < 0.95$. Within each month, PE is z-scored before the regression, and the lowest value reaches $z = -17.3$, so these few observations have high leverage. To inspect them, daily returns were rebuilt from the raw price files for 35 tickers: the 15 with the lowest minimum PE, which cover 97% of the sub-0.95 stock-months, and 20 tickers drawn at random. The rebuilt returns reproduce the cached PE exactly for all 2,189 stock-months. Three tie treatments were then compared: the original sort order, random tie-breaking (20 jittered draws), and dropping zero-return days.

The results are as follows (Figure 2). Ties are common in the low tail: 52% of its stock-months contain at least one tied four-day window, against 15% for the random tickers. Tie-breaking order still barely matters: the median PE change is zero in both groups. Of the 93 inspected stock-months below 0.95, only 4% rise above 0.95 under random tie-breaking, and the minimum moves from 0.805 to 0.817. The extreme case, SHREEASHTA at 2013-07-31, had 19.8% zero-return days, ties in 26% of its windows, and a median price of Rs 2.25 in the window. Dropping its zero days lowers PE further, to 0.681. Price is the common feature: 73% of the sub-0.95 stock-months have a window median price below Rs 10 (median Rs 3.55, against Rs 128 for the random tickers). With a fixed tick size, returns at such prices take only a few discrete values. The low tail therefore looks like genuine ordinal structure in very low-priced stocks, consistent with price discreteness or bid–ask bounce, rather than an artefact of the tie rule. This explanation was not tested further. One low-tail ticker (GBGLOBAL) shows window prices near

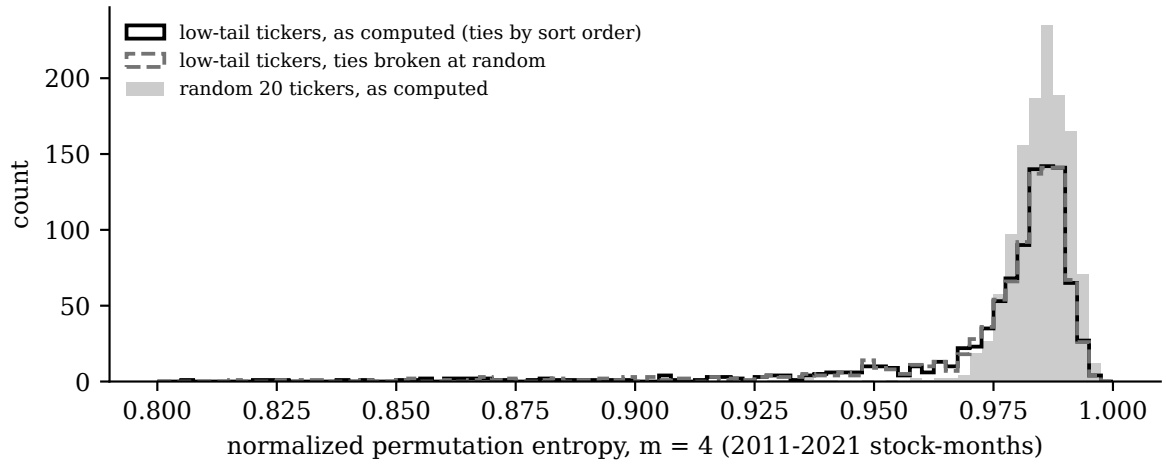


Figure 2: Revision diagnostic. PE_4 for 965 stock-months of the 15 low-tail tickers under the original tie rule (solid) and with ties broken at random (dashed), against 1,235 stock-months of 20 randomly drawn tickers (shaded), 2011–2021.

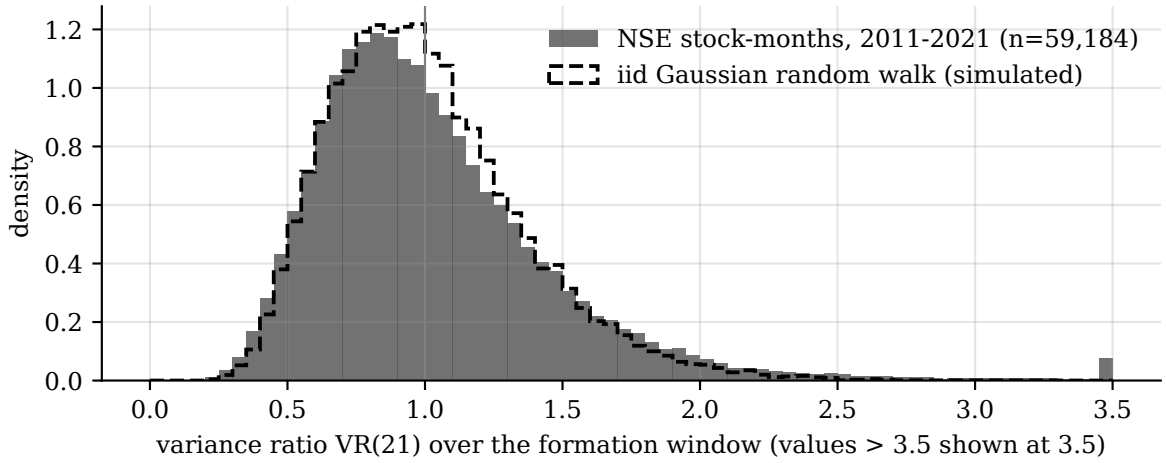


Figure 3: VR(21) for all Test D stock-months, 2011–2021 (bars), and 20,000 Gaussian random walks of length 232 passed through the same estimator (dashed).

Rs 88,600, which suggests a price-scale error in the raw file; it was not corrected.

Effect on the M7 gate. Re-estimating the full panel's tie treatment would need the full return panel, which was not available for this revision. Instead, the gate was rerun on the cached panel in two ways that remove the tail's leverage. Dropping stock-months with $PE_4 < 0.95$ gives gate t-statistics of -0.83 (2011–2026) and $+0.38$ (2011–2021). Replacing z(PE) with z(rank) gives -1.11 and $+0.47$. Every variant still fails the $|t| > 2$ rule.

5.3 Variance ratios and the random-walk probe

The observed median VR(21) is 0.950, and the simulated random-walk median is 0.958 (Figure 3). With 20,000 draws, the random-walk mean is 1.00 and the standard deviation 0.35 at length 232, and 1.00 and 0.34 at length 252. The source notes report a random-walk mean of 0.95 and standard deviation 0.30 at length 252 over 200 seeds, and describe the estimator as unbiased. The original simulation code was not found, so that figure could not be reproduced as the same

experiment. A reconstruction with Gaussian i.i.d. returns, length 252 and seeds 0–199 gives 0.949 and 0.297, matching it to two decimals. With 100 times as many draws, the mean is 1.00. The 0.95 is therefore consistent with Monte Carlo error over 200 draws (standard error about 0.02). It is not evidence that the estimator is unbiased, and that claim is withdrawn. Under the Gaussian random-walk null, sampling noise accounts for about 53% of the observed cross-sectional variance of VR (mean within-month standard deviation 0.47). Heavy tails and volatility clustering would raise that share.

5.4 Estimator stability and agreement

M7 measured rank stability as the Spearman correlation between a month's PE_4 ranking and the ranking recomputed on block-bootstrapped paths (20-day blocks, 50 resamples). The value was 0.440 over 12 months (range 0.411–0.477), against a bar of 0.6. Three of the 12 months fall in 2022 onward, and the nine development months average 0.444. PE_3 and PE_4 correlate at 0.770. In the notebook's check month, sample entropy correlated with PE_4 at only 0.06–0.11 across its (m, r) grid, so the two disorder measures rank stocks almost independently. The source note's statement that they "agree" is not supported. Volatility clustering is one possible reason why sample entropy departs from i.i.d. while PE does not; this is a hypothesis for future work.

6 What this paper does not establish

The paper does not show that path shape is uninformative in general. Only two estimators were tested, at one frequency (daily), over one window (232 rows), on one approximate universe. It does not test wavelet scale energy. It does not show that NSE returns are i.i.d.: the entropy diagnostic concerns ordinal patterns only, and sample entropy points the other way. It does not establish a zero effect, because effects smaller than the interval half-widths are not excluded. Nothing here validates out of sample: the entropy test used 2022 onward, and the restricted re-estimates were chosen after the full-sample result was known.

7 Limitations

- **Use of 2022 onward.** M7 read 2022–2026. A 2010–2021/2022–2026 split was already defined in earlier batch notebooks [6], but a code-level wall first appears in the Test C notebook (2026-08-08), after M7 (2026-08-04).
- **Gate 2 calibration (Test D).** The rule compared a p10–p90 range with a noise standard deviation of 0.30. A normal variable with that standard deviation already has a p10–p90 range of about 0.77, so the gate was not a like-for-like comparison. Future gates should use simulated null quantiles of the same statistic. Gate 2 never ran, so this flaw weakens the planned chain but does not change the gate-1 result.
- **Tie sensitivity of the full gate** was examined on a 35-ticker subset and through tail-robust re-estimates, not by rebuilding the full panel with an alternative tie rule.
- **Calendar defect and data quality.** The months listed in Section 2 await a calendar-corrected rerun. The 2025 volatility gap affects only the M7 full sample. The stale-price family of tickers has no filter yet.
- **Universe.** Membership comes from semi-annual snapshots, and the 2018H2 snapshot is truncated.

8 Conclusion

For the specified daily-return window and the two registered measures, neither permutation entropy nor the 21-day variance ratio passed its first forward-return gate. Most observed entropy

values lie near the finite-sample i.i.d. ordinal-pattern benchmark. The few that do not are concentrated in very low-priced stocks and are not an artefact of tie ordering. Simulated nulls suggest that sampling variation accounts for a substantial share of the cross-sectional dispersion in both measures. These diagnostics do not show that returns are i.i.d., or that every path-shape measure is uninformative. The result is limited to these estimators, this frequency and window, the approximate universe, and the sample. Path shape at lower frequencies, and path shape as a predictor of risk rather than return, remain untested.

9 Reproducibility

Table 3: Reproducibility summary. "Rev." marks revision diagnostics. Software for the local scripts: Python 3.13.16, NumPy 2.5.3, pandas 3.0.5, SciPy 1.18.1, matplotlib 3.11.2, pyarrow 26.0.0. Original notebooks ran in Google Colab (Python 3.10); their package versions were not recorded.

Item	Run	Settings
M7 gate	2026-08-04	rebalances 2011-01 to 2026-06; 84,144 stock-months (1,014 tickers); 176 FM months
M7 stability	2026-08-04	12 months; 20-day blocks; 50 resamples; seed 7
D gate	2026-08-08	2011-01 to 2021-12; 59,184 stock-months (865 tickers); 125 FM months
C gate / diagnostics	2026-08-08	2010-06 to 2021-12; 133 / 132 months
Intervals	drafting	NW lag 6; circular block bootstrap, block 6, 10,000 draws, seed 2026
PE / VR nulls	drafting	20,000 Gaussian series, $n = 232$, single `default_rng(7)` stream; SampEn null 500 draws
VR probe	rev.	`default_rng(s)`, $s = 0 \ldots 199$ and $0 \ldots 19{,}999$; $n \in \{232, 252\}$
Tie check	rev.	35 tickers, 2,189 stock-months; jitter $\pm 10^{-12}$, 20 draws, seed 20261014; random tickers drawn with the same seed

Scripts: `make_figures.py` (Figures 1–3, Table 2) and `revision_diagnostics.py` (Section 5.2, the strict Test D row, the VR probe, and the tail sensitivities). The revision diagnostics were run by the AI assistant from the author's archived files and checked against the original notebook outputs where those overlap.

Use of AI tools. Code for this project was written with the assistance of an AI coding assistant (Anthropic's Claude), working to specifications set by the author, and the manuscript was drafted with the same assistance. The author designed the research questions and decision rules and ran the main analyses. Some diagnostics added during revision were computed by the assistant from archived run outputs; these are labelled as revision diagnostics in the text. The author reviewed all code and text and takes full responsibility for the content.

Data and code availability. Code: https://github.com/mriganksahi-beep/India-Equity-Research/tree/main/p2_path_texture. The point-in-time panel construction is documented in https://github.com/mriganksahi-beep/India-Equity-Research/tree/main/data_pipeline. Price data from Yahoo Finance and index-membership snapshots from a licensed vendor are not redistributed; the repository contains the scripts to rebuild the panel.

A Related shelved gate: Test C (low-volatility anomaly)

Context only. Test C is not a path-texture test, has no forward-return statistic, and cannot support the conclusions above.

Test C asked whether a low-volatility sort would be tradeable in the liquid core, defined as the top 50% of names by trailing 21-day turnover, or whether low volatility simply proxies for liquidity. The

gate statistic was the mean monthly Spearman correlation between RV60 and Amihud illiquidity, over 133 months from 2010-06 to 2021-12 (2011-08, 2014-08, 2016-10, 2018-03, 2019-03 and 2020-11 are absent). The rule, written before the number was computed, was: proceed if the correlation is at most 0.20, shelve if it is at least 0.40, and surface for judgement in between. The full-universe correlation was 0.088 (95% block-bootstrap interval 0.056–0.118). The liquid-core correlation was 0.303 (0.261–0.343), which falls in the ambiguous band.

Two diagnostics followed. D1 checked shape: across liquid-core RV deciles, the mean Amihud rank rose steadily from 0.36 to 0.65. D2 checked size: the partial correlation given *share price*, a weak proxy for size, was 0.242 against a raw 0.303 (132 months), a 20% reduction. Both point to a real link between volatility and liquidity, so C was shelved.

On the chronology of D2: the earlier specification used a binary 50% cutoff, and the saved notebook uses graded bands. Colab metadata shows that the gate cells ran at about 06:15 UTC on 2026-08-08, and that the diagnostics cell was added in Colab and executed once (execution count 4) at 06:28 UTC. Its saved output was produced by the graded-band code, and the earlier saved version of the notebook contains no D2 code. Unrecorded runs cannot be excluded, so D1 and D2 are treated as exploratory diagnostics, not registered tests. Their verdict is the same under either rule (20% $<$ 50%). 2015-02 has only 56 names (a Budget Saturday session). The full universe shows a near-zero correlation while the liquid core shows a positive one, which is the opposite of what an "illiquid corner" explanation predicts. This is a descriptive fact, not a tested mechanism.

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